

Project Management

9. Resource Management

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Resource Management

Behind every successful project is a great team.

What is Resource Management

- Project Resource Management includes the processes to identify, acquire, and manage the resources needed for the successful completion of the project.
 - **Team resources** or personnel refer to the human resources.
 - **Physical resources:** equipment, materials, facilities, infrastructure
- Project team management:
 - Stakeholder management: focus on emotions, engagement, expectations, and influence.
 - Team members are seen as stakeholders, whose needs, expectations, and engagement must be understood and managed.
 - Team management: focus on roles, responsibilities, performance, and coordination
- The project manager should be ***both leader and manager*** of the project team.
 - A manager without leadership might meet deadlines but have a disengaged or burnt-out team.
 - A leader without management might inspire people but fail to deliver.

Leader Role

Focus: Vision, motivation, engagement, and influence

- **Inspire and motivate** the team toward project goals
- Create and **communicate a compelling vision**
- Build **trust and a positive team culture**
- Encourage **collaboration and innovation**
- Help the team **navigate change and uncertainty**
- Manage **team morale**, resolve interpersonal conflict



Manager Role

Focus: Planning, organizing, tracking, and controlling work

- Define **roles and responsibilities**
- Create and follow **project plans and schedules**
- Allocate **resources** efficiently
- **Monitor performance**, quality, and progress
- Ensure **task completion and deliverables**
- Enforce **project processes and standards**





Planning process group

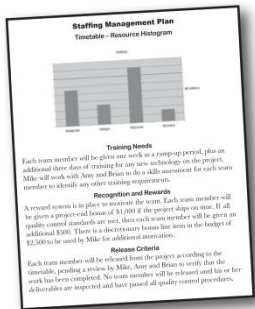
This is where you generate the estimates for the activities being performed.



Executing process group

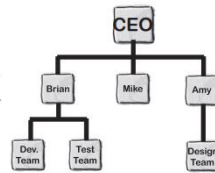
You'll need to train your team, and keep them motivated with rewards in order to get the project done.

No project goes perfectly. People have conflicts, and it's up to you to make sure they get resolved.



Plan Resource Management

Estimate Activity Resources



Acquire Resources

Once the project gets started, you need to put together your team. This means you may need to negotiate for some of the resources.



Develop Team

All three of these processes are part of the Executing process group, because they happen while your team's executing the project!



Manage Team



Control Resources



Monitoring & Controlling process group

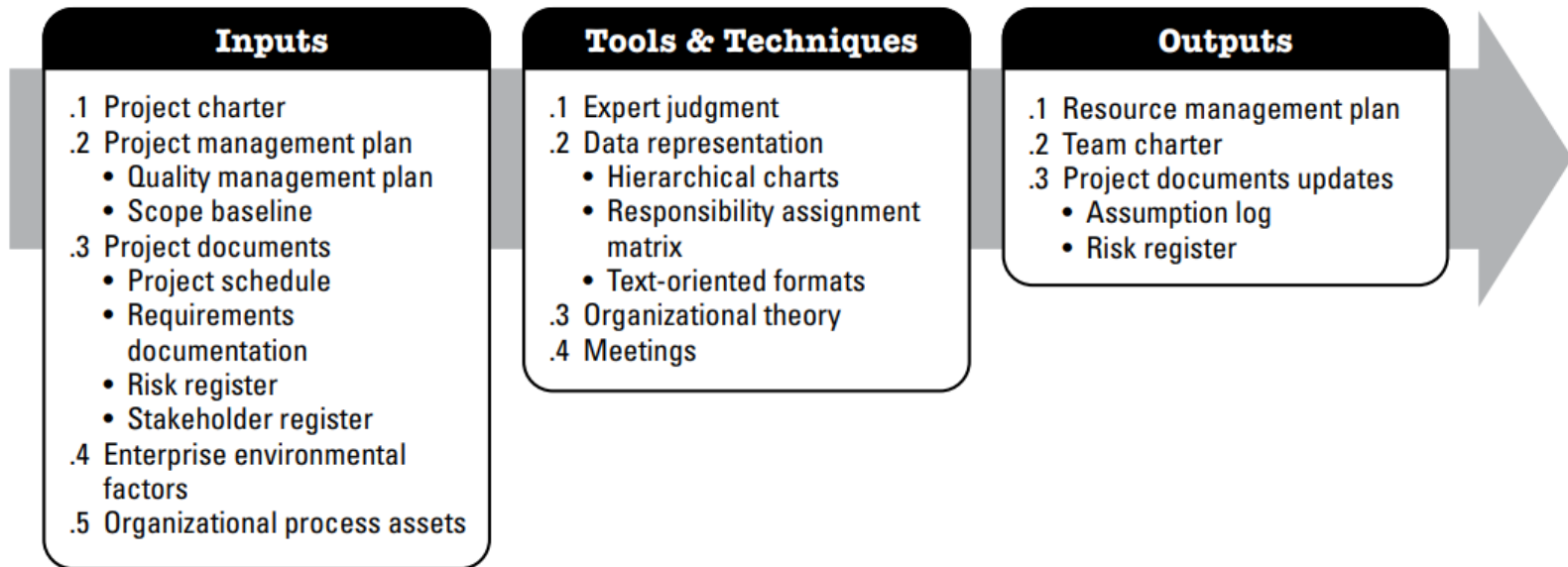
Never start a project unless all resources are available

Plan Resource Management

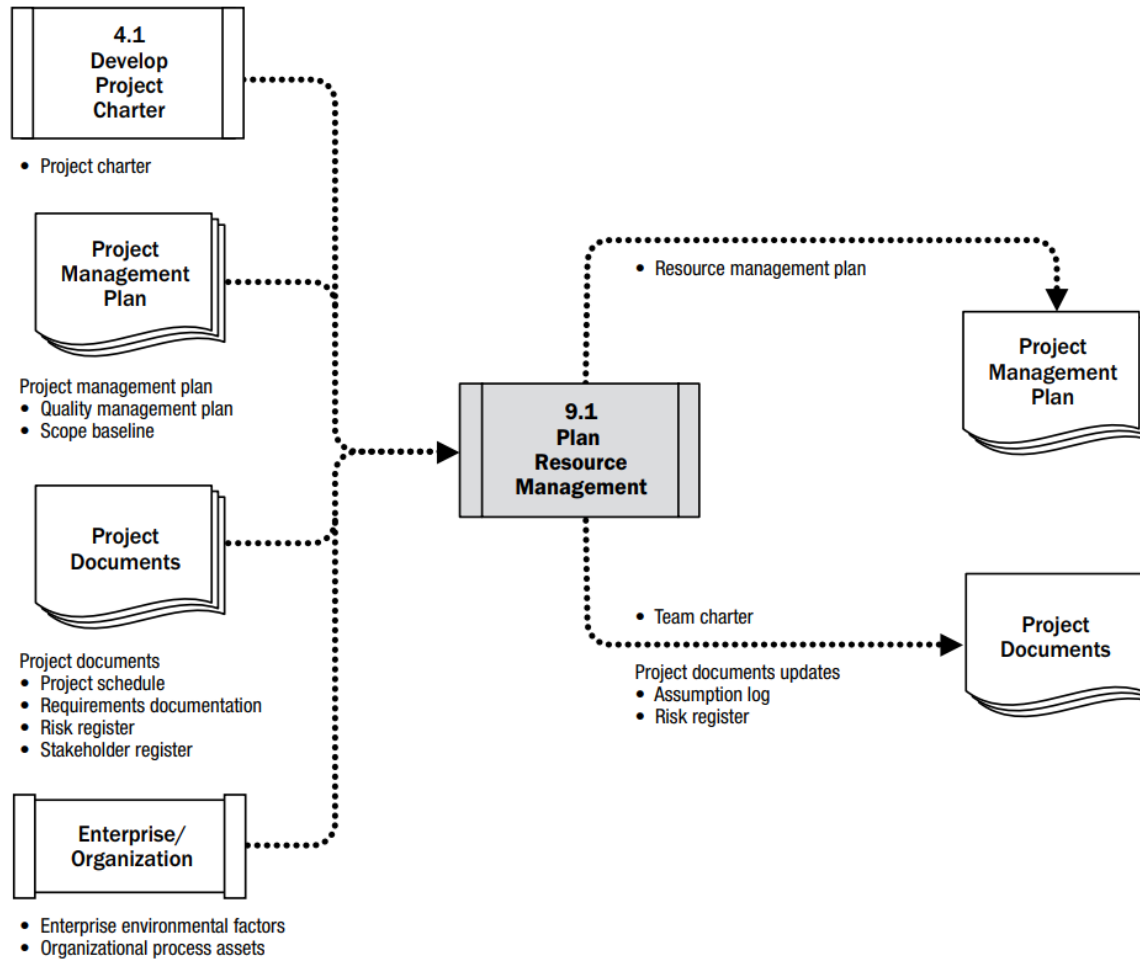
Plan Resource Management

- **What:** process of defining how to estimate, acquire, manage, and use team and physical resources.
 - Project resources may include team members, supplies, materials, equipment, services and facilities.
 - Resources can be obtained from the organization's internal assets or from outside the organization through a procurement process
- **Why:** determine and identify an approach to ensure that sufficient resources are available for the successful completion of the project.
 - Other projects may be competing for the same resources required for the project at the same time and location.
 - Significantly impact project costs, schedules, risks, quality, and other project areas.
- **When:** performed once or at predefined points in the project.

Process



Data Flow



Tools & Techniques

Expert judgment

- Expertise should be considered from individuals or groups with specialized knowledge or training in the resource management topics.

Meetings

- To allow for all project management team members to reach consensus on the resource management plan.



Tools & Techniques: Organizational Theory

Organizational Theory

- Organization theory provides ***frameworks*** and ***principles*** to understand how organizations are structured, how people behave in those structures, and how work and authority flow.
- Effective use of common techniques identified in organizational theory can ***shorten the amount of time, cost, and effort*** needed to create the Plan Resource Management.



Applying Organizational theories

Organization Theory	Core Concepts	Org Types	How It Helps in Resource Planning
Classical (Structural) Theory	Clear hierarchy, authority, specialization, division of labor	Functional Weak Matrix	Understand strict reporting lines, slow change, and rigid roles. Helps assign roles and manage availability.
Human Relations Theory	People are motivated by social needs, morale, and group belonging	Strong Matrix Projectized	Emphasizes team cohesion, morale, and leadership. Helps plan team culture and motivation techniques.
Contingency Theory	No one best way to organize—structure should depend on the situation	Any type, especially Matrix	Helps tailor the resource plan to fit specific organizational or project conditions (e.g., size, complexity, culture).
Systems Theory	Organizations are complex systems with interrelated parts	Matrix Functional	Encourages viewing resource planning in the context of dependencies between departments or projects.
Motivational Theories (e.g., Maslow, Herzberg)	Individual motivation influences performance	Projectized Agile organizations	Helps in planning non-monetary rewards, recognition, and work environment to boost team performance.
Theory X / Theory Y (McGregor)	X: People dislike work, need control Y: People seek responsibility, self-motivated	X: Functional Y: Agile, Projectized	In X-type, plan tight supervision; in Y-type, empower team with self-managed roles and responsibility sharing.

Applying Organizational theories

- **Theory X**

- Authoritarian Management Style
- People prefer to be directed and controlled. Workers are naturally lazy, selfish and hate work. Money is their only motivation.

- **Theory Y**

- Laissez-faire Management Style
- People prefer to work without supervision. Workers feel committed to their job, are responsible, self-directive, and exercise self-control

- **Maslow**

- Hierarchy of Needs



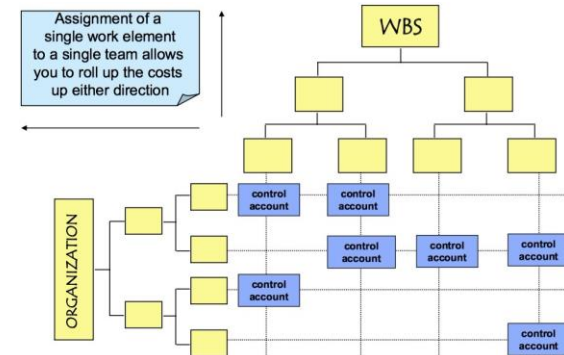
Maslow's hierarchy of needs

Tools & Techniques: Data representation

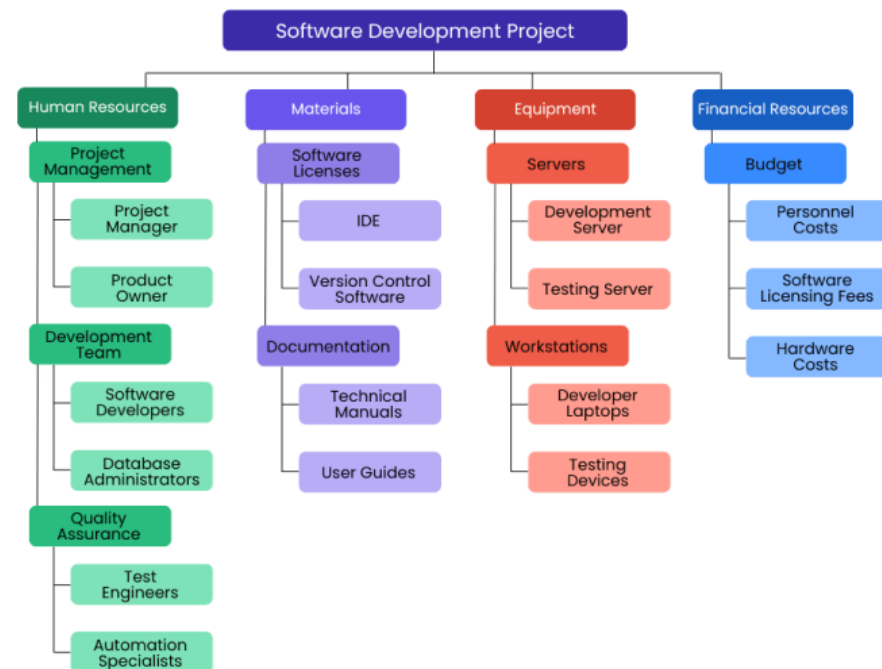
Hierarchical charts

Show positions and relationships in a graphical, top-down format.

- **Work breakdown structures (WBS):** showing high-level areas of responsibility
- **Organizational breakdown structure (OBS):** organization's existing departments, units, or teams, with the project activities or work packages listed under each department.
- **Resource breakdown structure.** hierarchical list of team and physical resources related by category and resource type that is used for planning, managing and controlling project work.



Resource Breakdown Structure



Tools & Techniques: Data representation

Responsibility Assignment Matrix (RAM) or RACI chart

Illustrate the connections between work packages, or activities, and project team members.

- **R-Responsible:** The person(s) who actually do the work to complete the task or deliverable.
- **A-Accountable:** The person who is ultimately answerable for the task's completion and success. One per task.
- **C-Consulted:** People who provide input or advice before the work is done. Usually subject matter experts.
- **I-Informed:** People who need to be kept up to date on progress or decisions but are not directly involved.

RACI Chart	Person				
Activity	Ann	Ben	Carlos	Dina	Ed
Create charter	A	R	I	I	I
Collect requirements	I	A	R	C	C
Submit change request	I	A	R	R	C
Develop test plan	A	C	I	I	R
R = Responsible A = Accountable C = Consult I = Inform					

Tools & Techniques: Data representation

Text-oriented format

- The documents are known by various names including position descriptions and role-responsibility-authority forms.
- These documents can be used as templates for future projects.

Roles, Responsibilities, and Authority

Role	Responsibility	Authority
1.	1.	1.
2.	2.	2.
3.	3.	3.
4.	4.	4.
5.	5.	5.
6.	6.	6.

Outputs: Resource Management Plan

Team resource management plan

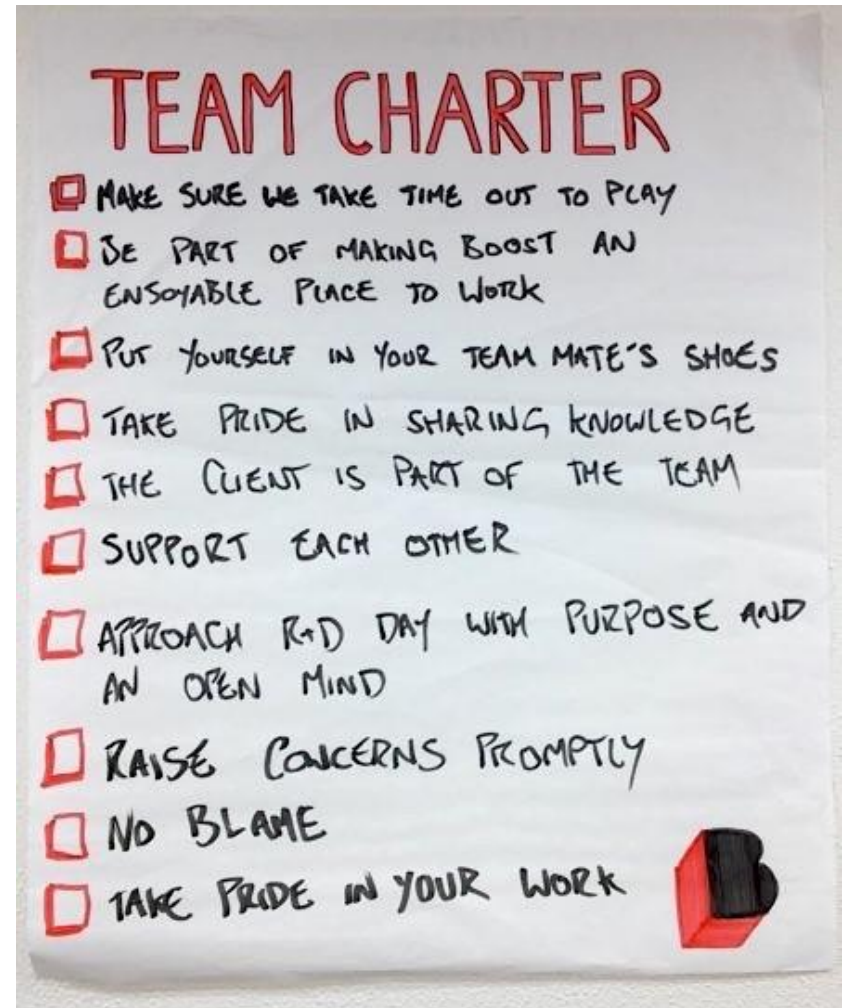
- Guidance on how project team resources should be defined, staffed, managed, and eventually released.
- Include but not to be limited to the following:
 - **Project organization chart**
 - Roles & responsibilities
 - Authority & competences
 - Resources identification
 - Team acquisition
 - Team training
 - Team development
 - Recognition and reward
 - Team release

Physical resource management plan

- Guidance on how physical resources should be identified, categorized, allocated, managed, controlled and released.
- Include but not to be limited to the following:
 - **Resource breakdown structure**
 - Resource identification
 - Resource acquisition
 - Resource allocation
 - Resource control
 - Resource release

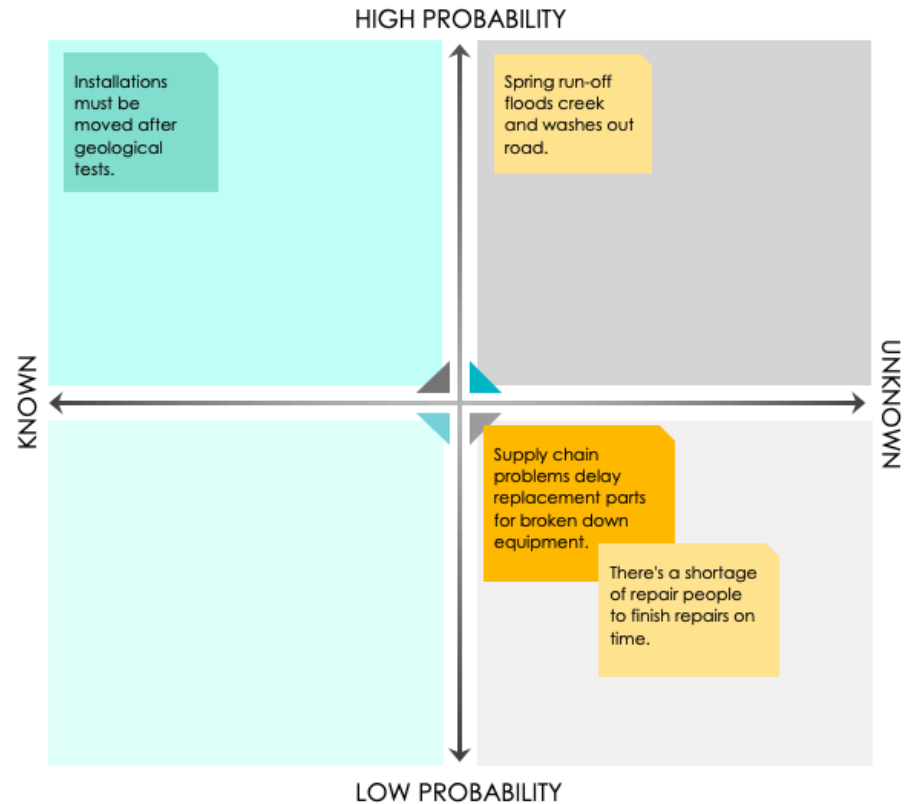
Outputs: Team charter

- The team charter establishes **clear expectations** regarding acceptable behavior by project team members.
- The team charter works best when the team develops it, or at least has an opportunity to **contribute to it**.
- All project team members **share responsibility** for ensuring the rules documented in the team charter are followed.
- The team charter can be reviewed and updated **periodically** to ensure a continued understanding of the team ground rules and to orient and integrate new team members.



Outputs: Project documents updates

- **Assumption log.** Assumptions regarding the availability, logistics requirements, and location of physical resources as well as the skill sets and availability of team resources.
- **Risk register.** Risks associated with team and physical resource availability or other known resource-related risks.



Inputs

Project charter

- **The key stakeholder list**
- **Summary milestones**
- Pre-approved financial resources

Project management plan

- **Scope management plan:** the deliverables that drive the types and quantities of resources that will need to be managed.
- **Quality management plan:** helps define the level of resources

Project documents

- **Stakholder register**
- Requirement documents
- Project schedule
- Risk register

Enterprise environmental factors

- Organizational structure and culture
- Existing human resources
- Personnel policies
- Marketplace condition

Organizational process assets

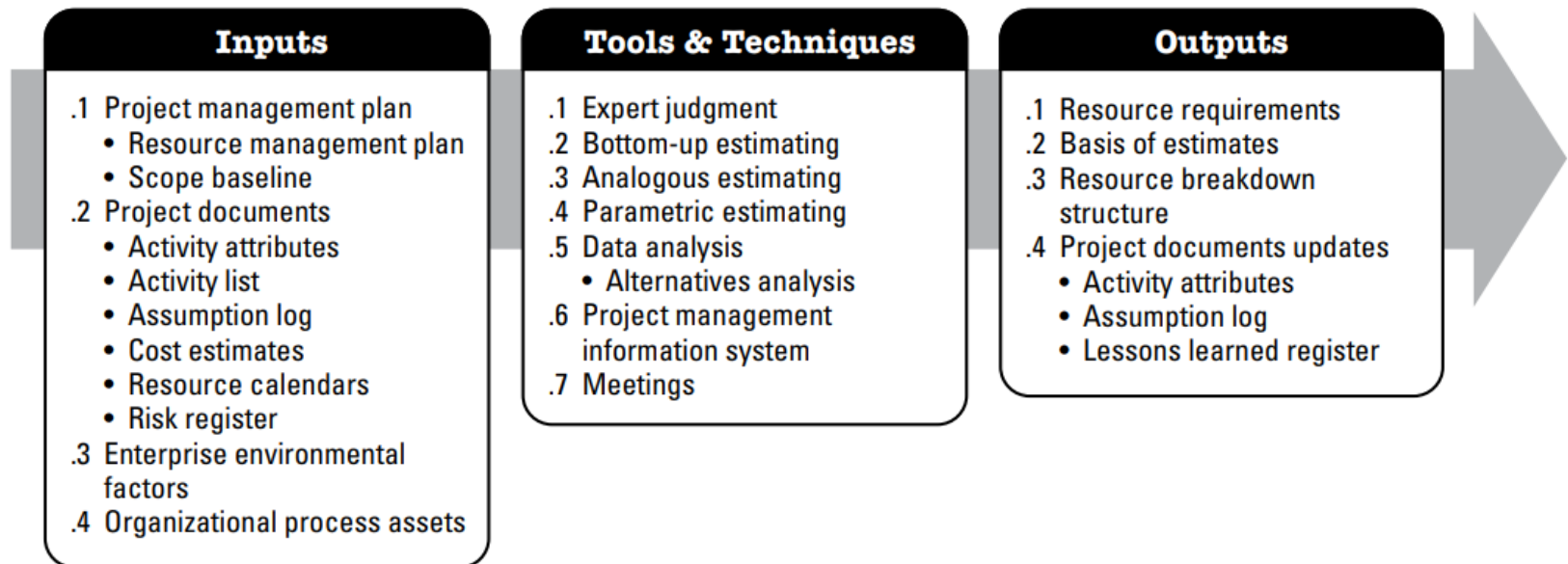
- Policies, and role descriptions;
- Templates for org. charts and position descriptions
- Escalation procedures for handling issues
- Historical information

Estimate Activity Resource

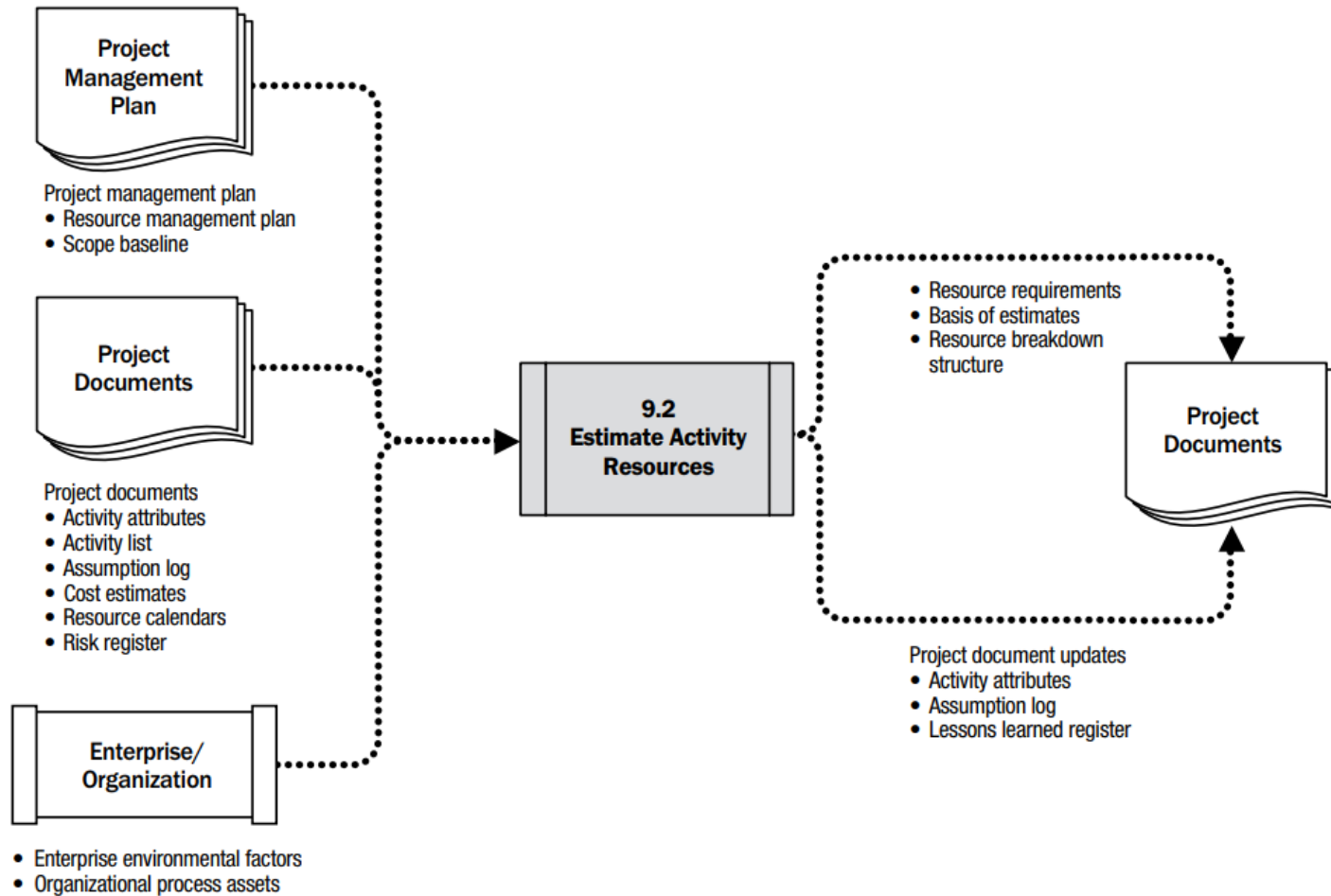
Estimate Activity Resource

- **What:** process of estimating team resources and the type and quantities of materials, equipment, and supplies necessary to perform project work.
 - Estimate Activity Resources process is closely coordinated with other processes, such as the Estimate Costs process.
- **Why:** identifies the type, quantity, and characteristics of resources required to complete the project.
- **When:** performed periodically throughout the project as needed.

Process



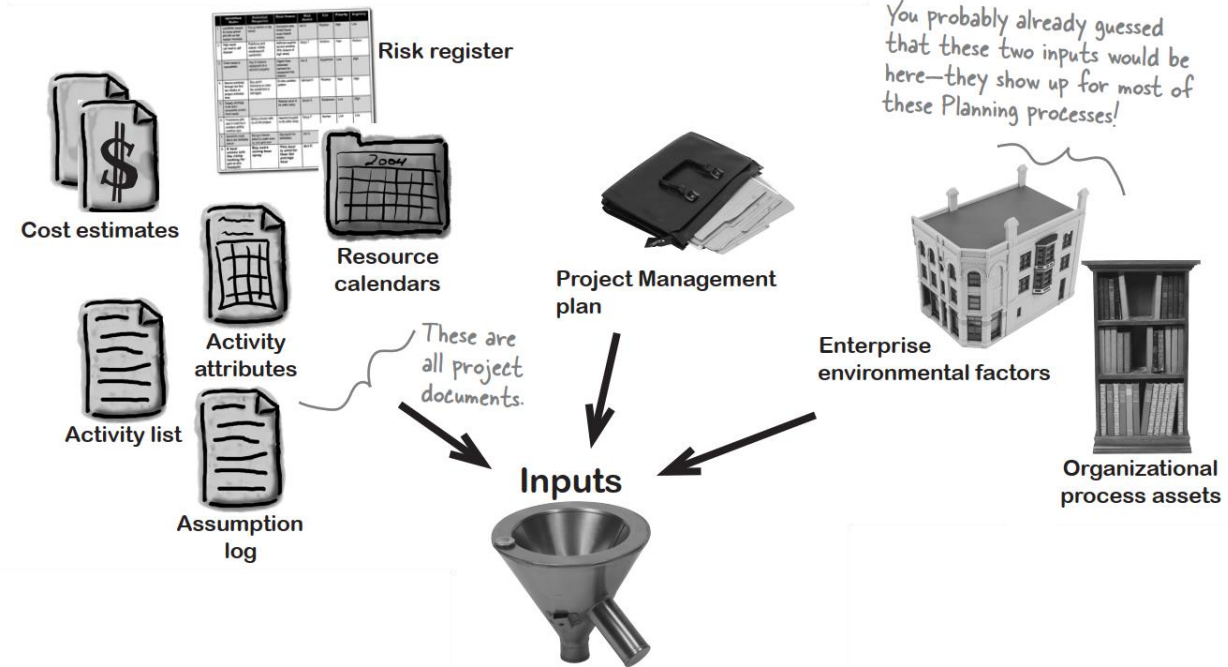
Data Flow



Inputs

Resource calendars

A resource calendar identifies the working days, shifts, start and end of normal business hours, weekends, and public holidays **when each specific resource is available.**



Tools & Techniques

Analogous estimation is making estimates based on the number of resources and amount of effort necessary to complete a project like the one you're currently estimating.

Parametric estimation is when estimates are made by typing values into a mathematical model that has been created based on data from past projects

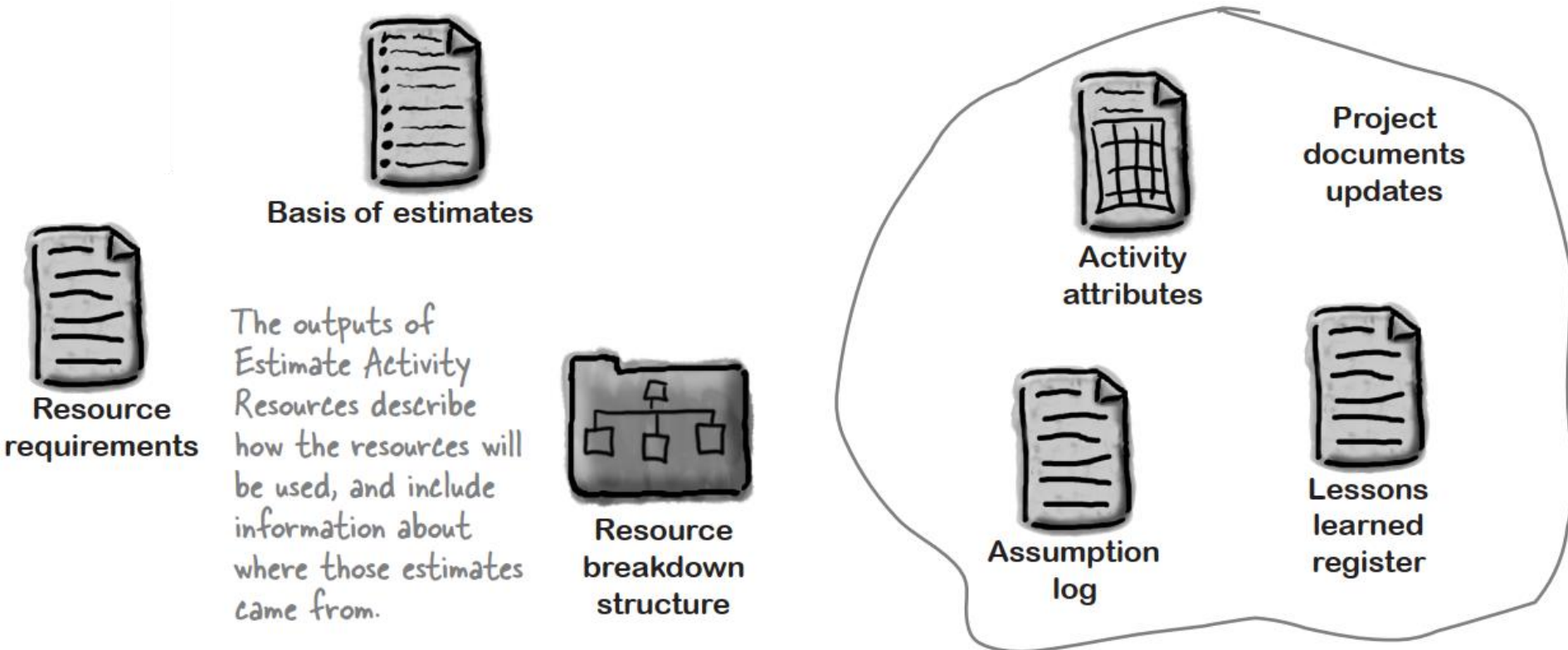
Bottom-up estimating is a technique that you may have used before without even knowing it! It means breaking down complex activities into pieces, and working out the resource assignments for each of those simpler pieces using the other six tools and techniques.

Expert judgment means bringing in experts who have done this sort of work before and getting their opinions on what resources are needed.

Project management information system like Microsoft Project will often have features designed to help project managers play around with resources and constraints and find the best combination of assignments for the project.

Meetings are used to get stakeholders to work collaboratively to estimate the project's resource needs.

Outputs



RESOURCE REQUIREMENTS				
Project Title: _____			Date Prepared: _____	
ID	Resource	Quantity	Assumptions	Comments

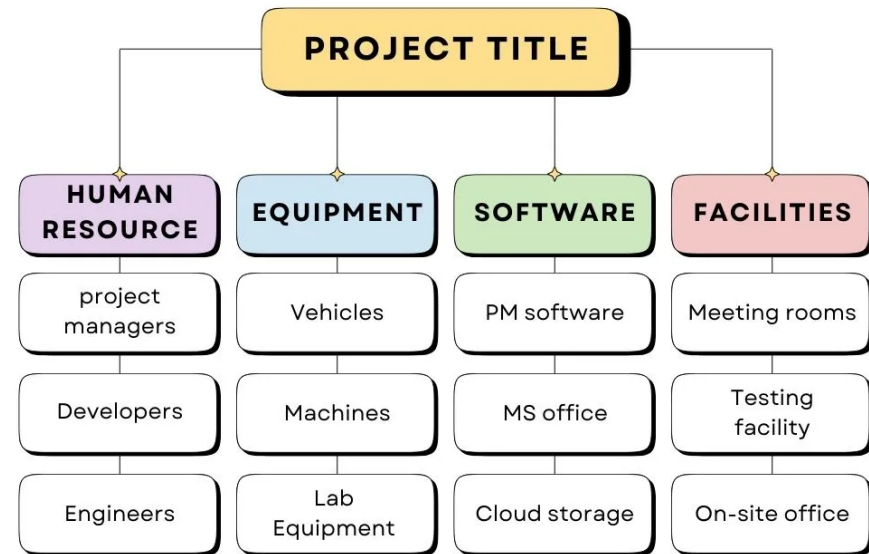
Page 1 of 1

Outputs

Resource Breakdown Structure

A hierarchical structure of the identified resources by resource category and resource type .

- E.g. of resource categories - labor, material, equipment , supplies etc
- Resource Types can include the skill level, grade level, etc
- **In Plan Resource Management**, the resource breakdown structure was used to guide the categorization for the project. In this process it is a completed document that will be used to acquire and monitor resources.

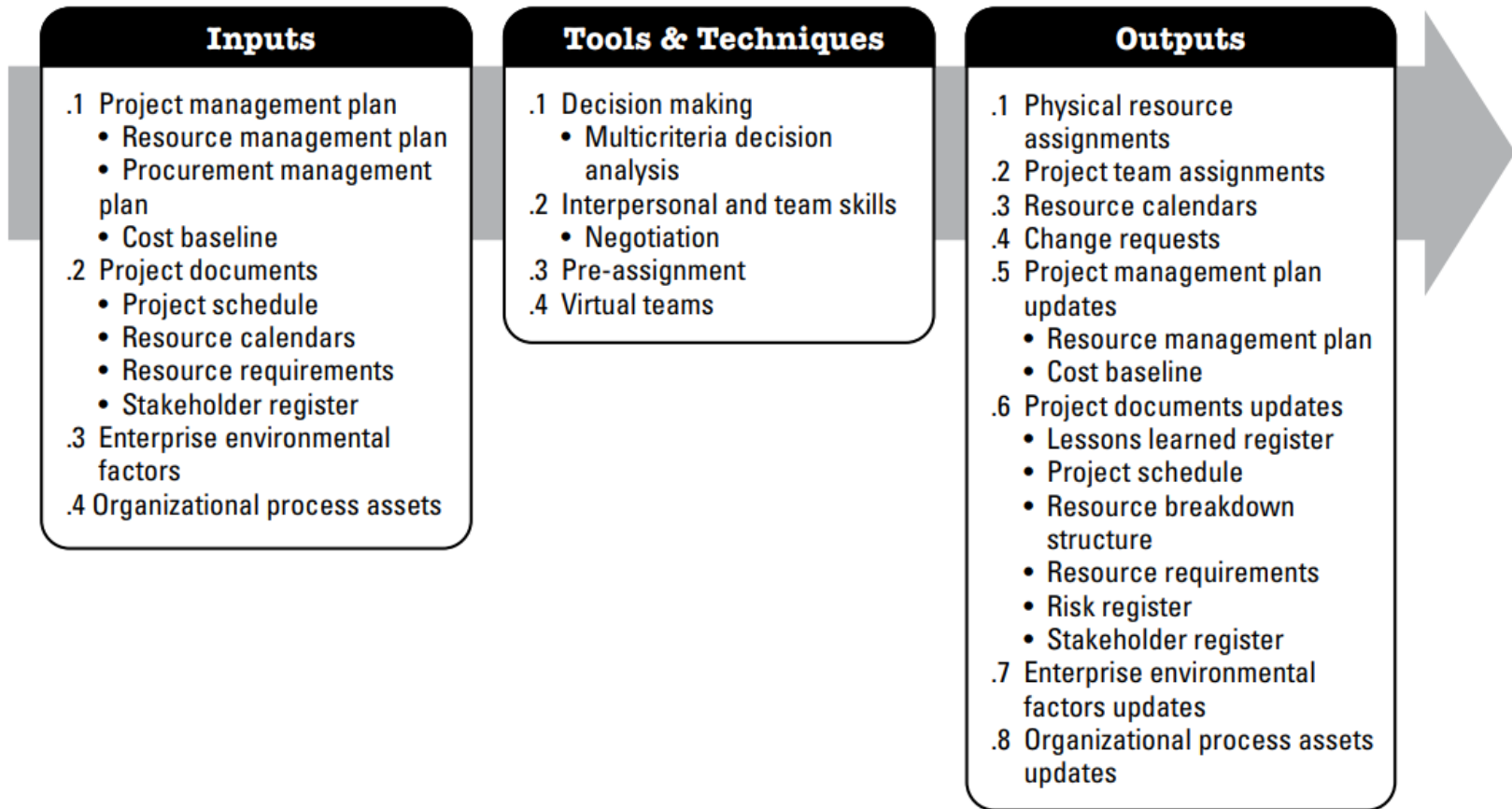


Acquire Resources

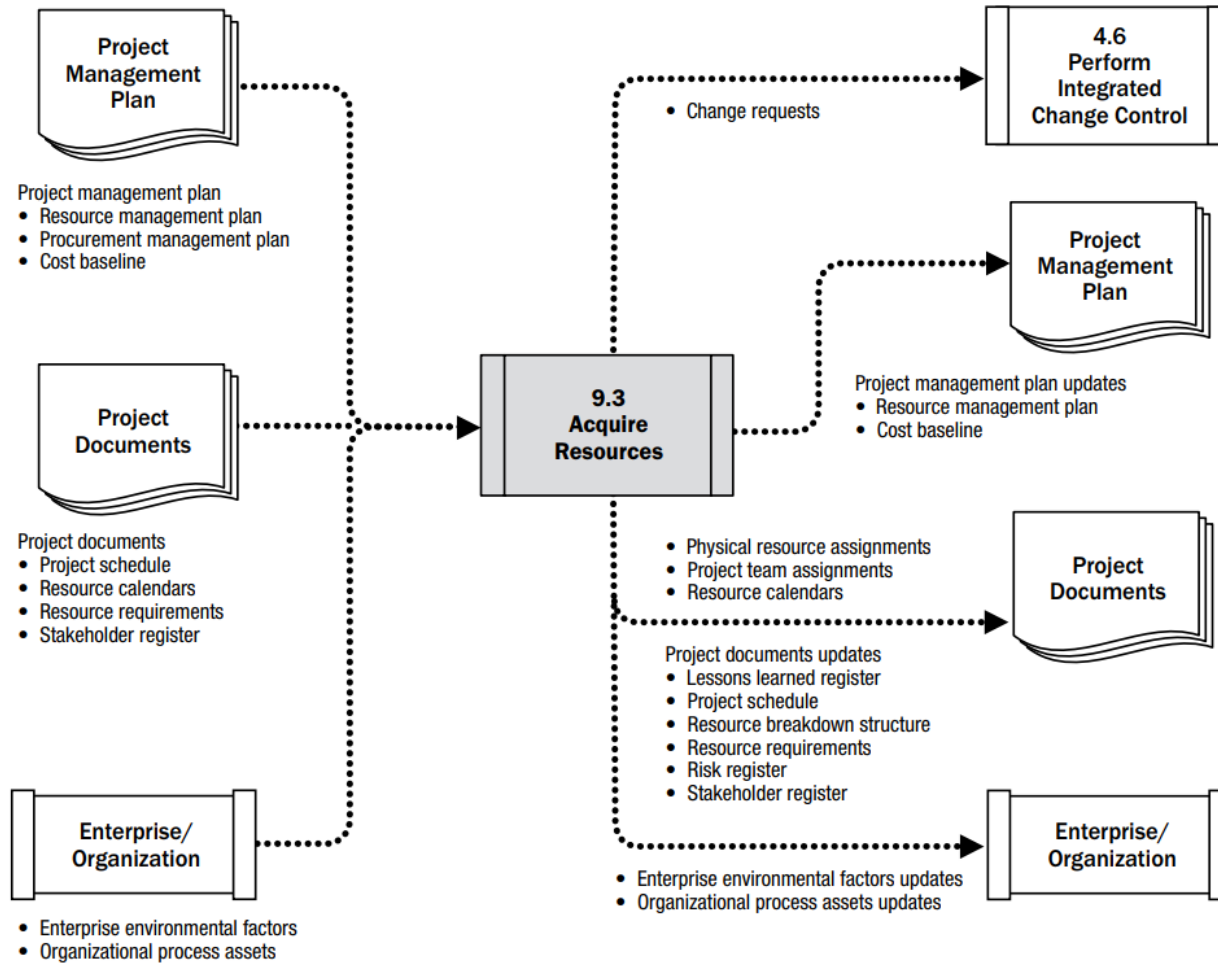
Acquire Resources

- **What:** process of obtaining team members, facilities, equipment, materials, supplies, and other resources necessary to complete project work.
 - Internal resources are acquired (assigned) from functional or resource managers.
 - External resources are acquired through the procurement processes.
- **Why:** The project management team may or may not have direct control over resource selection because of collective bargaining agreements, use of subcontractor personnel, a matrix project environment, internal or external reporting relationships, or other reasons.
 - *Insufficient resources* or capabilities decrease the probability of success and, in a worst-case scenario, could result in project cancellation
- **When:** performed periodically throughout the project as needed.

Process



Data Flow



Tools & Techniques

Decision making

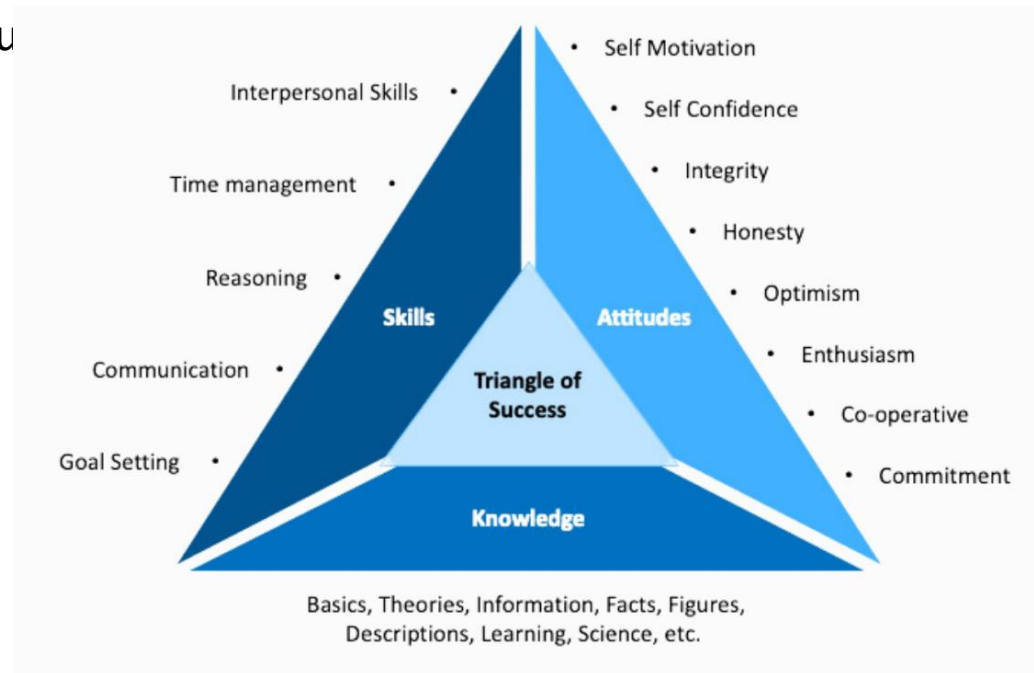
Multi-criteria decision analysis is often used to select physical project resources or the project team.

ASK/ASKE model:

- Attitude
- Skills
- Knowledge
- Experience
- International factors.

ACA model

- Availability
- Cost
- Ability



Build a secure API for user authentication, scheduled over 3 weeks.

Candidate	Availability	Cost	Ability
Dev A (Internal)	Fully available for 3 weeks	\$3,000/week	High – 5+ years in Spring Boot, knows project context
Dev B (Contractor)	Part-time (50% over 3 weeks)	\$2,000/week	Medium – 3 years in Spring Boot, no internal experience
Dev C (New hire)	Available after 1 week delay	\$2,500/week	Low/Unknown – Onboarding required, unfamiliar with tech stack

Candidate	Availability Score (1–3)	Cost Score (1–3)	Ability Score (1–3)	Total	Notes
Dev A	3 – Fully available	2 – Higher cost	3 – High ability	8	Most suitable despite higher cost
Dev B	2 – Half-time	3 – Lowest cost	2 – Medium ability	7	May delay progress unless scope reduced
Dev C	1 – Delay in availability	2 – Mid-level cost	1 – Low/unknown ability	4	Risky for critical path; not recommended

Tools & Techniques

Negotiation

Many projects need to negotiate for required resources:

- **Functional manager** to ensure availability of component staff.
- **Other PM teams** to assign scarce/specialized resources.
- **External organizations and suppliers.** Provide appropriate, scarce, specialized, qualified, certified, or other specific team or physical resources.



Tools & Techniques

Pre-assignment

Sometimes you can actually build the assignments into your Resource Management plan. Sometimes you have resources who are guaranteed to you when you start the project, so you don't need to negotiate for them.

Virtual teams

Your team members may not all work in the same location. This is really useful when you're relying on consultants and contractors for outsourced work. Instead of meeting in person, they'll use phone, email, instant messaging, and online collaboration tools to work together.



Outputs

Project Team Assignments

Records the team members and their roles and responsibilities for the project (may use RACI chart, org chart).

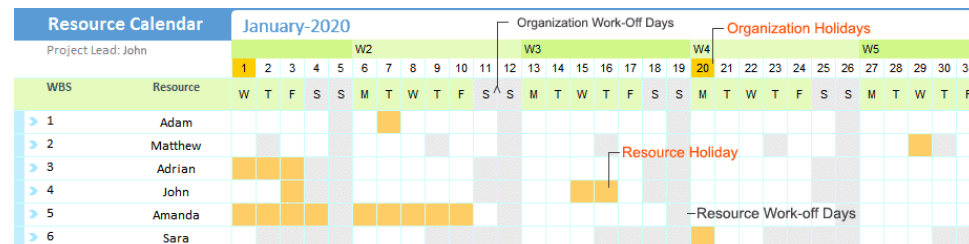
Team Member	Role	Assigned Tasks
Alice	Backend Developer	User authentication API
Bob	UI/UX Designer	Mobile app interface mockup
Carol	QA Tester	Regression testing after Sprint 2

Physical Resource Assignments

Records the material, equipment, supplies, locations, and other physical resources that will be used during the project

Resource Calendar

A resource calendar identifies the working days, shifts, start and end of normal business hours, weekends, and public holidays when each specific resource is available.



Outputs

Project management plan updates

- **Resource management plan.**
- **Cost baseline.**

The update is always happened because:

- People seldom fit the exact staffing requirements that are planned.
- Promotions, retirements, illness, performance issues, and changing work loads....

Project documents updates

- **Stakeholder register.**
- **Project schedule.** Changes to the project schedule
- **Resource breakdown structure.**
- **Resource requirements.**
- **Lessons learned register.**
- **Risk register.**

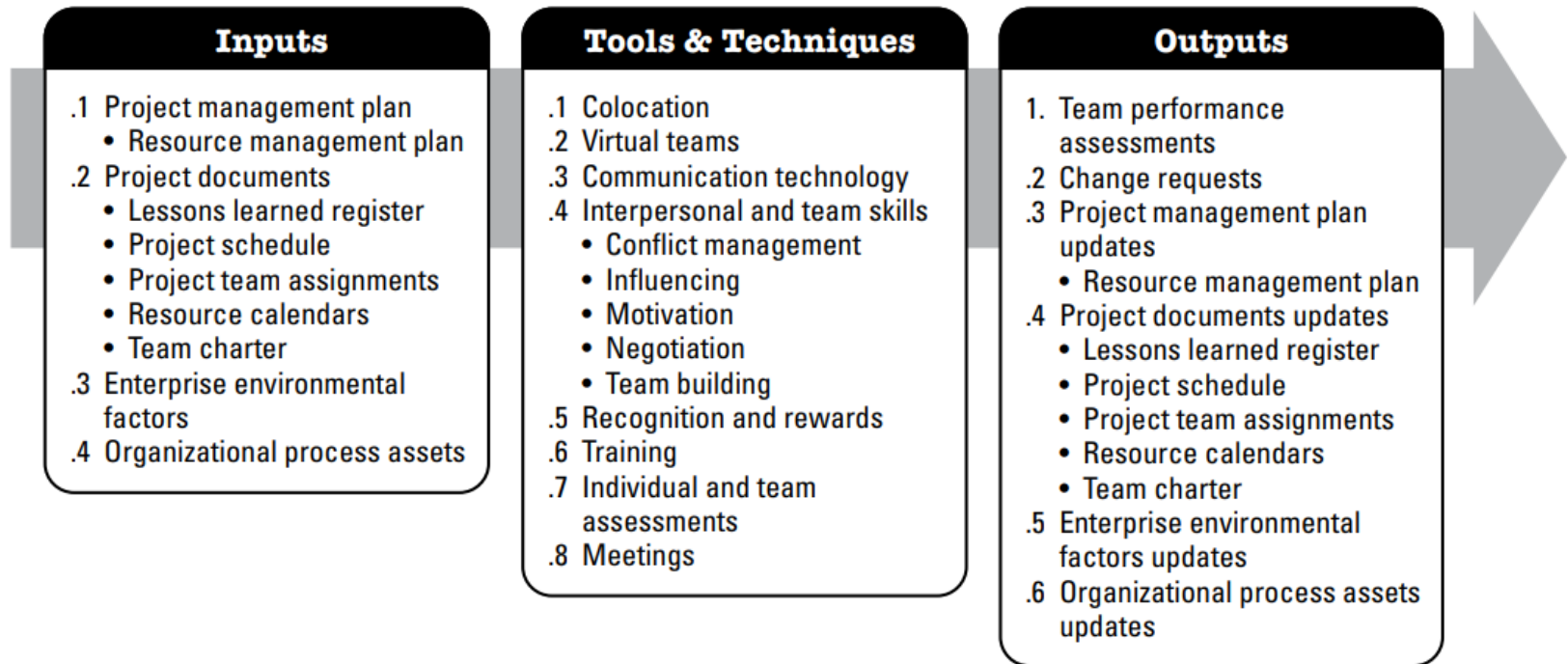
Develop Team

*Teamwork is a **critical factor** for project success, and developing effective project teams is one of the primary responsibilities of the project manager.*

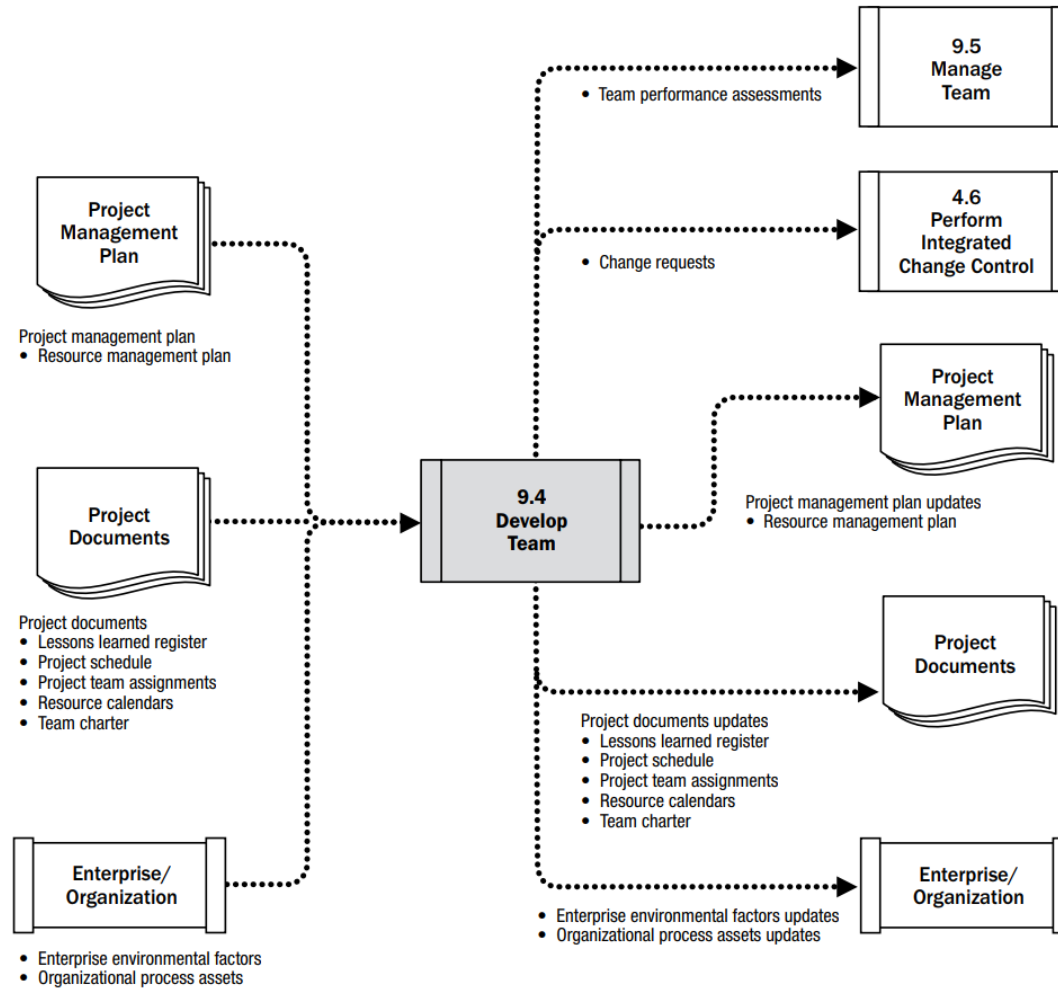
Develop Team

- **What:** process of improving competencies, team member interaction, and the overall team environment to enhance project performance.
- **Why:**
 - improved teamwork
 - enhanced interpersonal skills and competencies
 - motivated employees
 - reduced attrition
 - improved overall project performance.
- **When:** performed throughout the project.

Process



Data Flow



PM's responsibility

Motivation

- One of your most important jobs as project manager is keeping the team motivated and constantly monitoring them to make sure they stay motivated.
- A really effective way to motivate your team is to set up a reward system. But make sure that they understand exactly what they're being rewarded for—and it *must* be fair, or it could backfire!
- Training is another great way to keep a team motivated. When people feel that they're growing professionally, they stay more involved and get more excited by their work.

Management

- When the project is being planned, you're directing everything—but by the time it's executing, the project manager is more of a coach and a facilitator.
- That's why it's really important for a project manager to have “soft skills”—you need to really understand what makes your team members tick, and help with their problems.
- A really good way to make sure that your project team sticks together is to establish **ground rules** for your project, which set a standard for how everyone works together.

5 stages of Team development

Forming

Team acquaints and establishes ground rules. Formalities are preserved and members are treated as strangers.



Storming

Members start to communicate their feelings but still view themselves as individuals rather than part of the team. They resist control by group leaders and show hostility.



Norming

People feel part of the team and realize that they can achieve work if they accept other viewpoints.



Performing

The team works in an open and trusting atmosphere where flexibility is the key and hierarchy is of little importance.



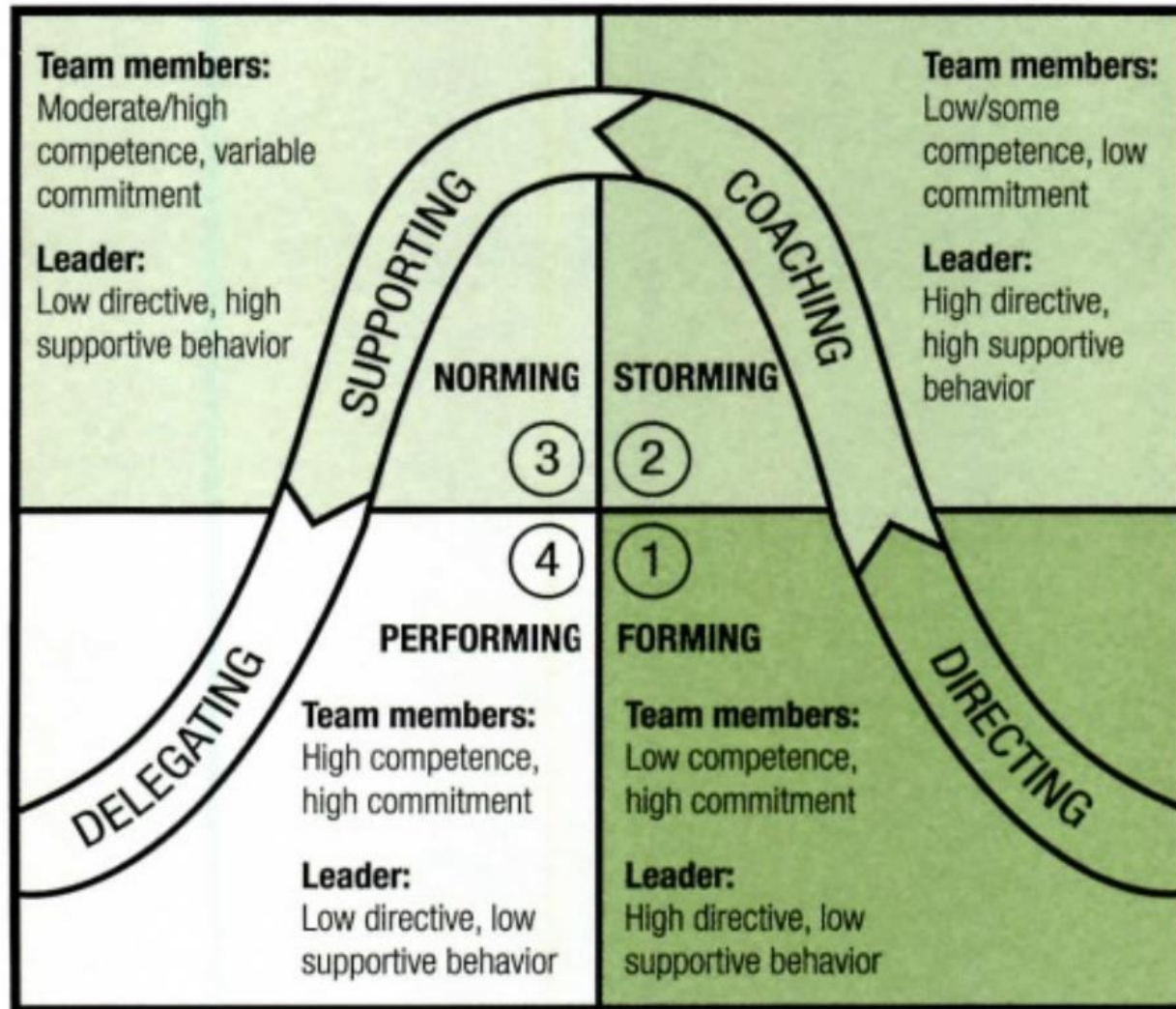
Adjourning

The team conducts an assessment of the year and implements a plan for transitioning roles and recognizing members' contributions.



Tuckman ladder

Leadership styles and stages of Team



Tools & Techniques

Colocation is the opposite of virtual teams. When you have all of your team located in the same room, you can increase communication and help them build a sense of community. Sometimes that room is called a **war room**

Virtual teams are teams that sit in different offices and use technology to communicate with one another.



Interpersonal and team skills

Leadership

Giving the team a goal to shoot for and helping them to see the value in the work they are doing, let the team:

- know the end product that they're building
- **understand the value that that product is going to bring to the company.**

=> A project manager needs to constantly **remind the team of the vision** they're working toward, and make decisions to help keep the team on track toward it.



Team building

A PM need to help the team:

- learn to depend on and trust one another.
- understanding about how they'll communicate and stay motivated when things go wrong.

If you're open about your decision-making processes and communicate often about what you're doing, you can **help your team to bond.**



Interpersonal and team skills

Motivation demonstrates to your team the value that the project has for them.

- Making sure that people are compensated and rewarded financially for their work.
- The team also needs to know how the tasks they're doing contribute to project success, and what's in it for them.

Motivating your team is about **helping them to be satisfied with the job they're doing**, recognizing them when they do a good job, and keeping them challenged with new and different problems.



Communication

It's not enough to make the best decisions to get your project done; you've got to make sure that everybody in the team knows why you're making them, and feels like you're being **open and honest** about what's motivating every decision you make. If the people on your team feel like they're always getting the information they need from you and that they're never in the dark, they'll be able to trust you and one another more.



Interpersonal and team skills

Influencing is all about using your relationships with the people on your team to get them to cooperate in making good decisions for the project. When you lead by example, you **show your team how you want them to behave by doing it**. It may seem subtle, but the way you work as a project manager can **set the standard for your teammates**.



Negotiation helps the people on your team come to an agreement about how to work together. It's important when you're negotiating to **listen to both parties** and to make sure that you **make it clear when concessions are made**. That should get everyone to see both sides of the issue and know that you're negotiating a fair resolution to it.



Interpersonal and team skills

Political and cultural awareness

means knowing the people on your team and understanding their backgrounds. Since projects sometimes span more than one culture, it's important to take the time to **understand the similarities and differences in the working environments** across the project team. It's equally important to communicate with your team members and understand what motivates them.

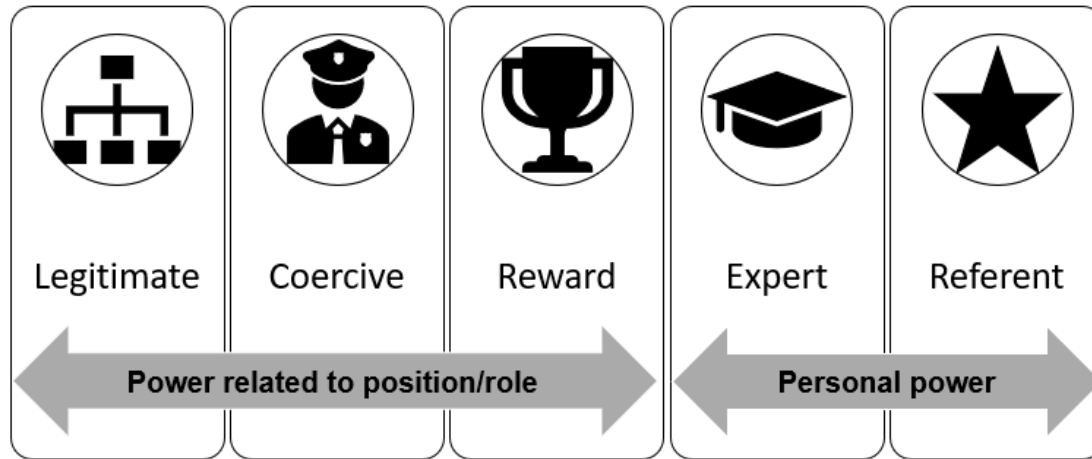


Trust building, coaching, and conflict management are also important in managing the interpersonal dynamics on your team.

CONFLICT RESOLUTION



5 forms of power



- 1 Legitimate Power**
From the position in the hierarchy
- 2 Expert Power**
From having expertise in a particular area
- 3 Referent Power**
From interpersonal relationships
- 4 Coercive Power**
From the ability to influence through threats
- 5 Reward Power**
From the ability to offer and influence incentives

*The most effective forms of power are **reward power**, where the project manager sets up rewards and recognition for the team, and **expert power**, which means the team respects the project manager's technical expertise*

Match the power form

Legitimate power

A project manager sets up an "Accolade Wall" and posts awards for team members who come in ahead of schedule.

Reward power

Everyone always does what Shelly says because everyone knows that senior management loves her.

Expert power

A functional manager assigns a tester to work on the project manager's team.

Referent power

The programmers always listen to the team lead because he's a really good software architect.

Motivate your team

Theory X

- People prefer to be directed and controlled. Workers are naturally lazy, selfish and hate work. Money is their only motivation.
- Authoritarian Management Style

Theory Y

- People prefer to work without supervision. Workers feel committed to their job, are responsible, self-directive, and exercise self-control.
- Laissez-faire Management Style

Maslow

People have needs, and until the lower ones are satisfied they won't even begin to think about the higher ones.

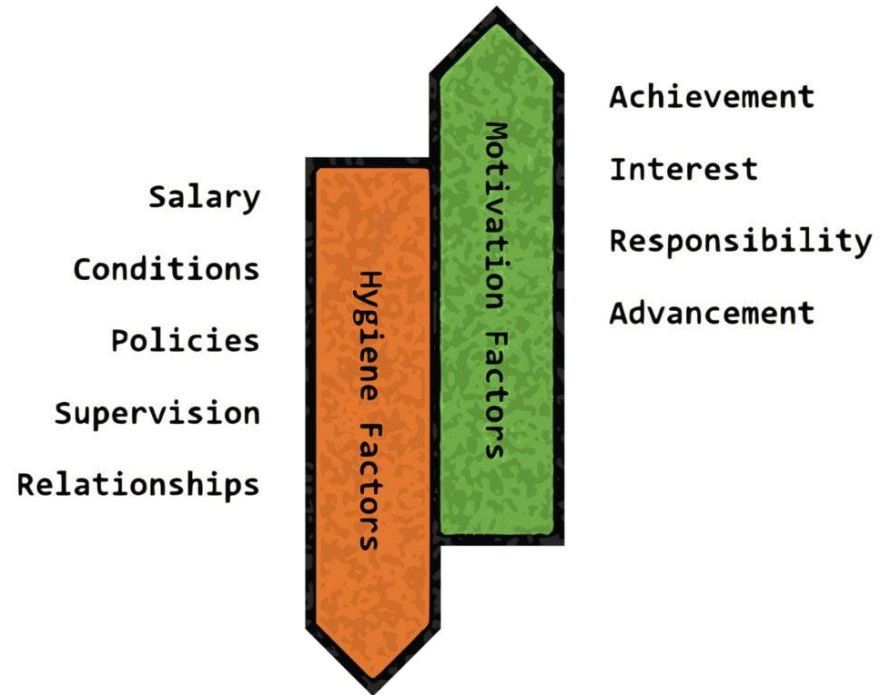


Maslow's hierarchy of needs

Motivation-Hygiene Theory

Herzberg says that people *need the stuff they normally expect* out of a job—like hot coffee—**before** you can get them motivated about achievement and personal growth.

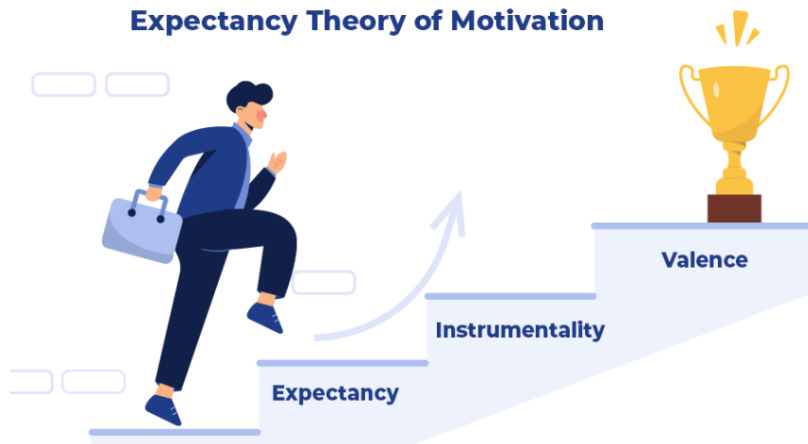
- **Motivators** (or satisfiers) are intrinsic to the job itself and lead to job satisfaction when present.
- **Hygiene factors** (or dissatisfiers) are extrinsic to the work and, while not directly causing satisfaction, can lead to dissatisfaction when absent.



Motivate your team

Expectancy Theory says that you need to give people an expectation of a reward in order to motivate them—but this works only if that **award is achievable**. If everyone knows the award is either worthless or impossible to achieve, it will actually demotivate them!

McClelland's Achievement Theory says that people need to be motivated. Achievement is when someone performs well and is **recognized** for it. Power means he or she has a lot of control or influence in the company. And someone feels a strong sense of affiliation from being a part of a working team and having good relationships with coworkers.



Recognition and Rewards

Win lose (zero sum) rewards

That only a limited number of project team members can achieve such as team members of the month, can hurt team cohesiveness.



Win-win rewards

Behavior that every one can achieve such as turning in progress reports on time, tends to increase support among team members.



Tools & Techniques

Training

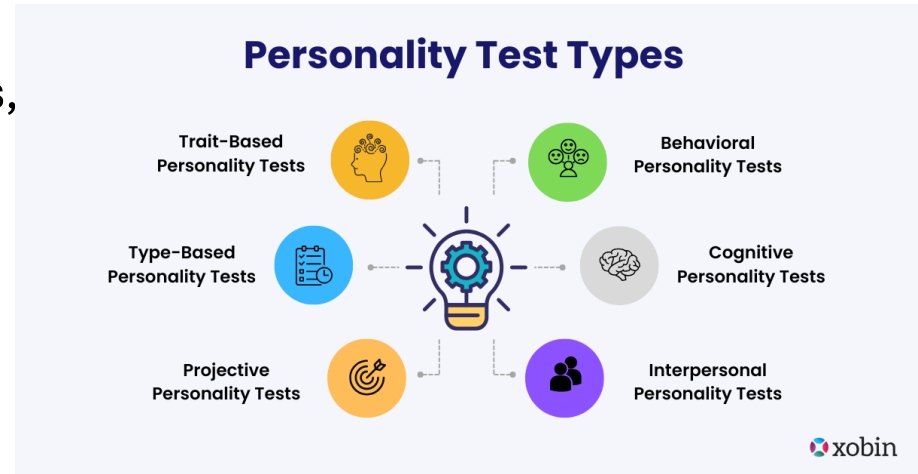
- Enhance the competencies of the project team members.
- Training costs could be included in the project budget or supported by the performing organization if the added skills may be useful for future projects.
- It can be formal or informal
- It may be performed by in-house or by external trainers.



Tools & Techniques

Individual and team assessment tools

- These tools help to assess team members' preferences, aspirations, **how they process and organize information, how they make decisions, and how they interact with people.**
- Various tools are available: attitudinal surveys, specific assessments, structured interviews, ability tests, and focus groups



Outputs

Team performance assessments

- Formal or informal assessments of the project team's effectiveness.
- As a result of conducting an evaluation of the **team's overall performance**, the project management team can identify the specific training, coaching, mentoring, assistance, or changes required to improve the team's performance.
- This should also include identifying the appropriate or required resources necessary to achieve and implement the improvements identified in the assessment.

TEAM PERFORMANCE ASSESSMENT			
Project Title: _____		Date Prepared: _____	
Technical Performance			
Scope	☐ Exceeds Expectations	☐ Meets Expectations	☐ Needs Improvement
Comments:			
Quality	☐ Exceeds Expectations	☐ Meets Expectations	☐ Needs Improvement
Comments:			
Schedule	☐ Exceeds Expectations	☐ Meets Expectations	☐ Needs Improvement
Comments:			
Cost	☐ Exceeds Expectations	☐ Meets Expectations	☐ Needs Improvement
Comments:			
Page 1 of 3			

Outputs

Team performance assessments

- Formal or informal assessments of the project team's effectiveness.
- As a result of conducting an evaluation of the **team's overall performance**, the project management team can identify the specific training, coaching, mentoring, assistance, or changes required to improve the team's performance.
- This should also include identifying the appropriate or required resources necessary to achieve and implement the improvements identified in the assessment.

TEAM PERFORMANCE ASSESSMENT			
Project Title: _____		Date Prepared: _____	
Technical Performance			
Scope	☐ Exceeds Expectations	☐ Meets Expectations	☐ Needs Improvement
Comments:			
Quality	☐ Exceeds Expectations	☐ Meets Expectations	☐ Needs Improvement
Comments:			
Schedule	☐ Exceeds Expectations	☐ Meets Expectations	☐ Needs Improvement
Comments:			
Cost	☐ Exceeds Expectations	☐ Meets Expectations	☐ Needs Improvement
Comments:			
Page 1 of 3			

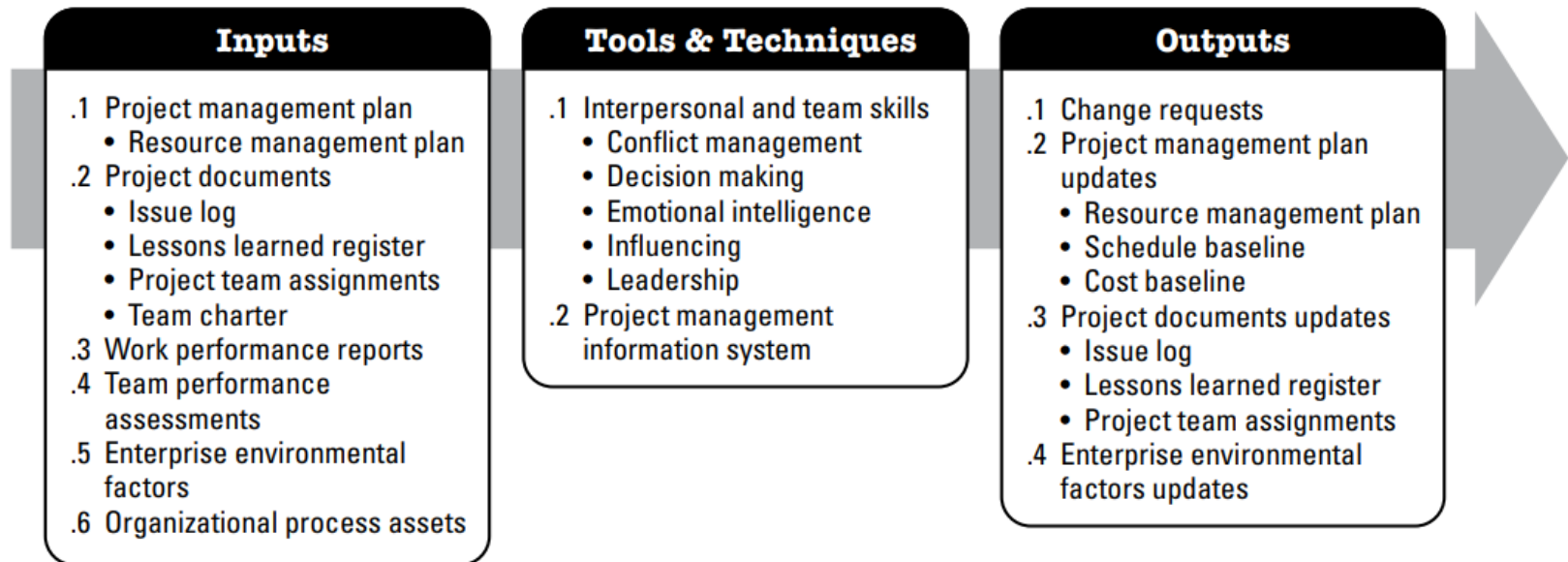
Manage Team

*Teamwork is a **critical factor** for project success, and developing effective project teams is one of the primary responsibilities of the project manager.*

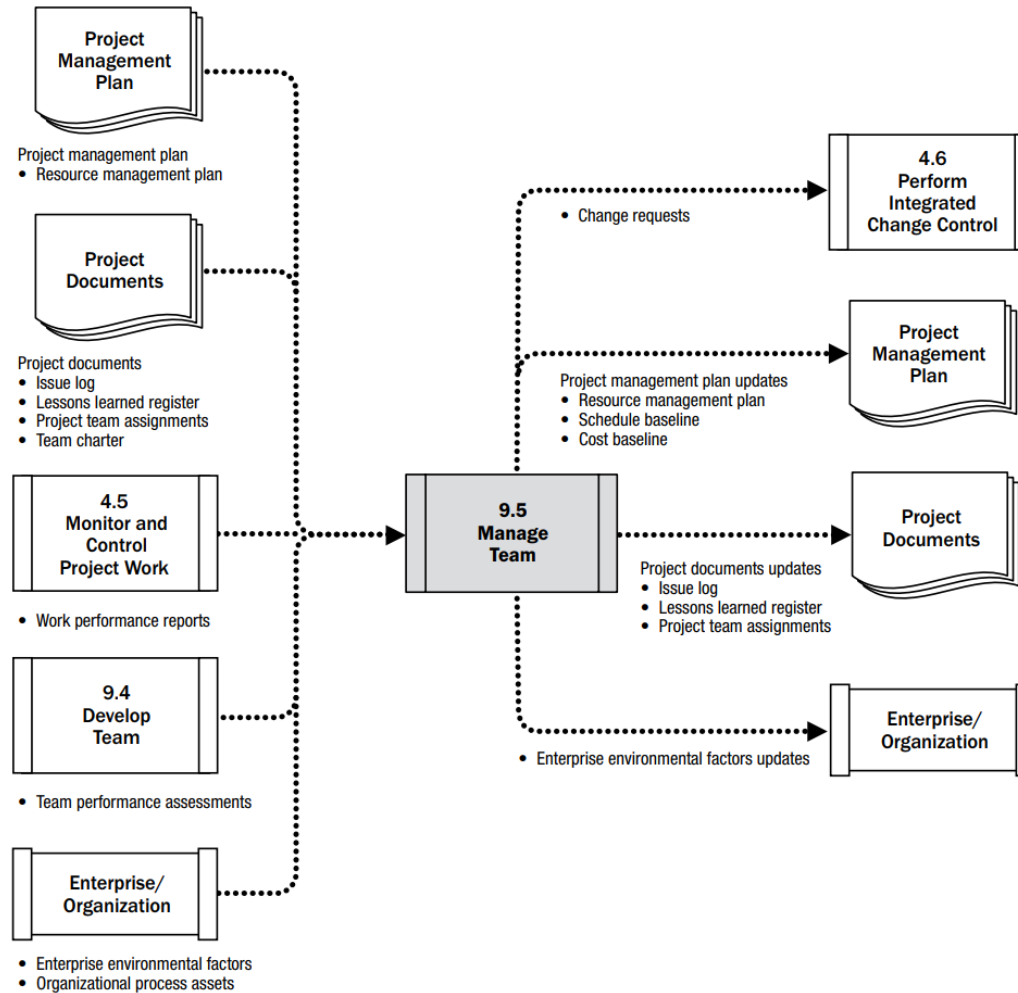
Manage Team

- **What:** process of tracking team member performance, providing feedback, resolving issues, and managing team changes to optimize project performance.
 - Team management involves a combination of skills with special emphasis on communication, conflict management, negotiation, and leadership.
- **Why:** influences team behavior, manages conflict, and resolves issues.
- **When:** performed throughout the project.
- **How:**
 - Requires a variety of management and leadership skills for fostering teamwork and integrating the efforts of team members to create high-performance teams.
 - The project manager needs to be sensitive to both the willingness and the ability of team members to perform their work and adjust their management and leadership styles accordingly.

Process



Data Flow



Tools & Techniques

Conflict Management

Sources of conflict include:

- Scarce resources
- Schedule: decide who gets what, when
- Priorities: one project or person is more important than another
- Personal work styles
- Cost: especially where contracts are involved.
- Technical opinions

Conflict Resolution

How to reduce:

- Team ground rules, group norms
- Solid project management practices (communication planning, role definition)

The best way to resolve a conflict is to collaborate with others: do your research, figure out what's behind it, and fix the root cause.

Conflict Resolution

1



AVOIDING

(Low assertiveness, Low cooperativeness)

Sidesteps or ignores conflicts

Best for: Minor issues or when temporary delay is needed

Example: Postponing decisions on non-critical system updates

2



COMPETING

(High assertiveness, Low cooperativeness)

Pursues own concerns over others

Best for: Quick, decisive action needed

Example: Critical security implementations

3



ACCOMMODATING

(Low assertiveness, High cooperativeness)

Yields to others' needs

Best for: Maintaining relationships

Example: Flexible work schedules

4



COMPROMISING

(Moderate assertiveness, Moderate cooperativeness)

Finds middle ground

Best for: Equal power situations

Example: Budget allocation

5



COLLABORATING

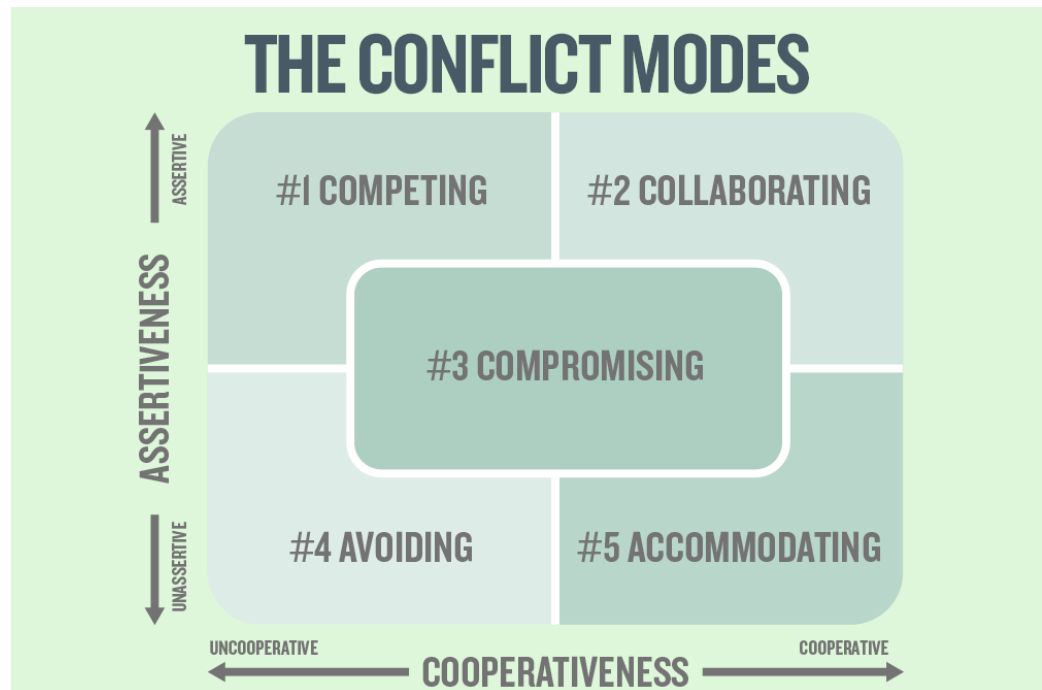
(High assertiveness, High cooperativeness)

Seeks win-win solutions

Best for: Complex issues requiring integration

Example: Technology stack decisions

THE CONFLICT MODES

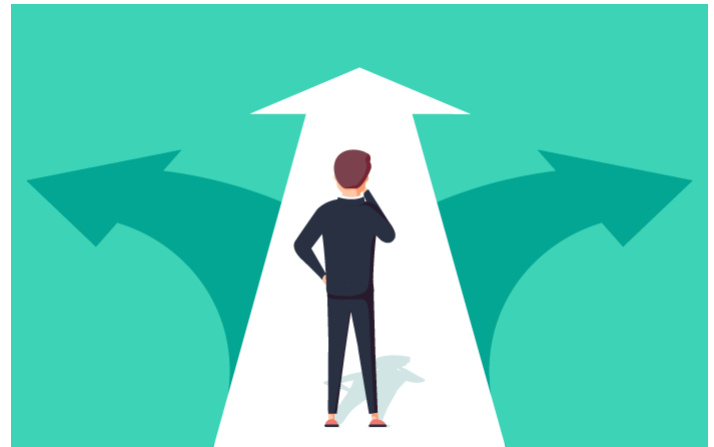


Tools & Techniques

Decision making

Decision making, in this context, involves the ability to negotiate and influence the organization and the project management team, rather than the set of tools described in the decision making tool set.

- Define the goals to be served
- Follow a decision-making process
- Ensure team members are engaged and focused on project goals
- Stimulate team creativity
- Get agreement and support for team's decision



Tools & Techniques

Leadership

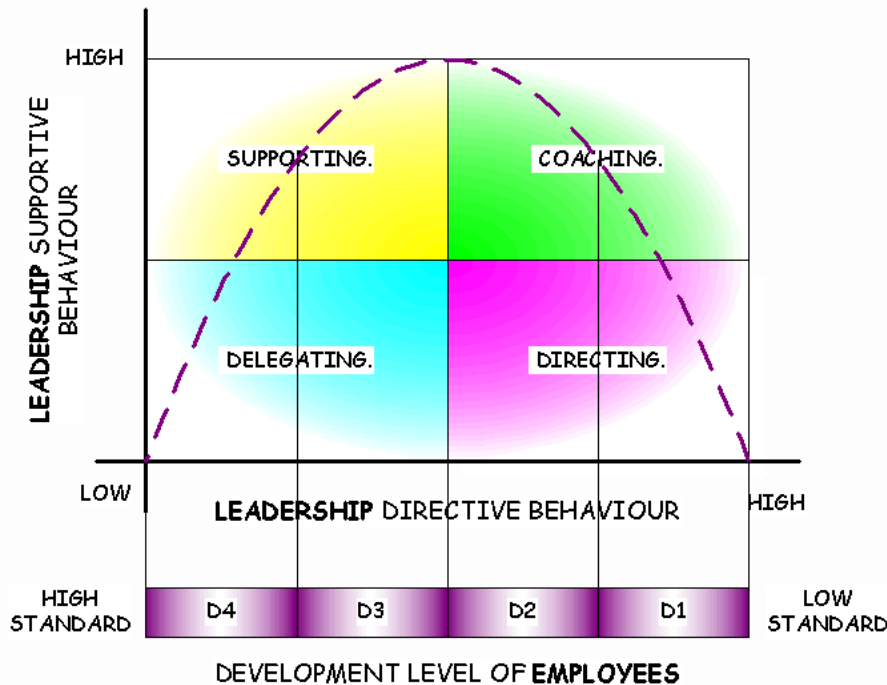
Leadership is the ability to lead a team and inspire them to do their jobs well.

- Successful projects require leaders with strong leadership skills.
- It is especially important to communicate the **vision and inspire** the project team to achieve high performance.



Leadership

Situational Leadership
Hersey-Blanchard model



The right leadership style will depend on the person or group being led

1. **Directing:** This style involves telling others what to do
2. **Coaching:** In coaching, the manager helps others achieve their goals.
3. **Supporting:** the project manager provides assistance along the way.
4. **Delegating:** the leader is still involved in decisions; however, the process and responsibility has been passed to the individual or group. The leader stays involved to monitor progress.

Tools & Techniques

Emotional Intelligence

Emotional intelligence is the ability to identify, assess, and manage the personal emotions of oneself and other people, as well as the collective emotions of groups of people.

The team can use emotional intelligence to reduce tension and increase cooperation by identifying, assessing, and controlling the sentiments of project team members, anticipating their actions, acknowledging their concerns, and following up on their issues.

Influencing

PMs often have little or no direct authority over team members in a matrix environment, their ability to influence stakeholders on a timely basis is critical to project success.

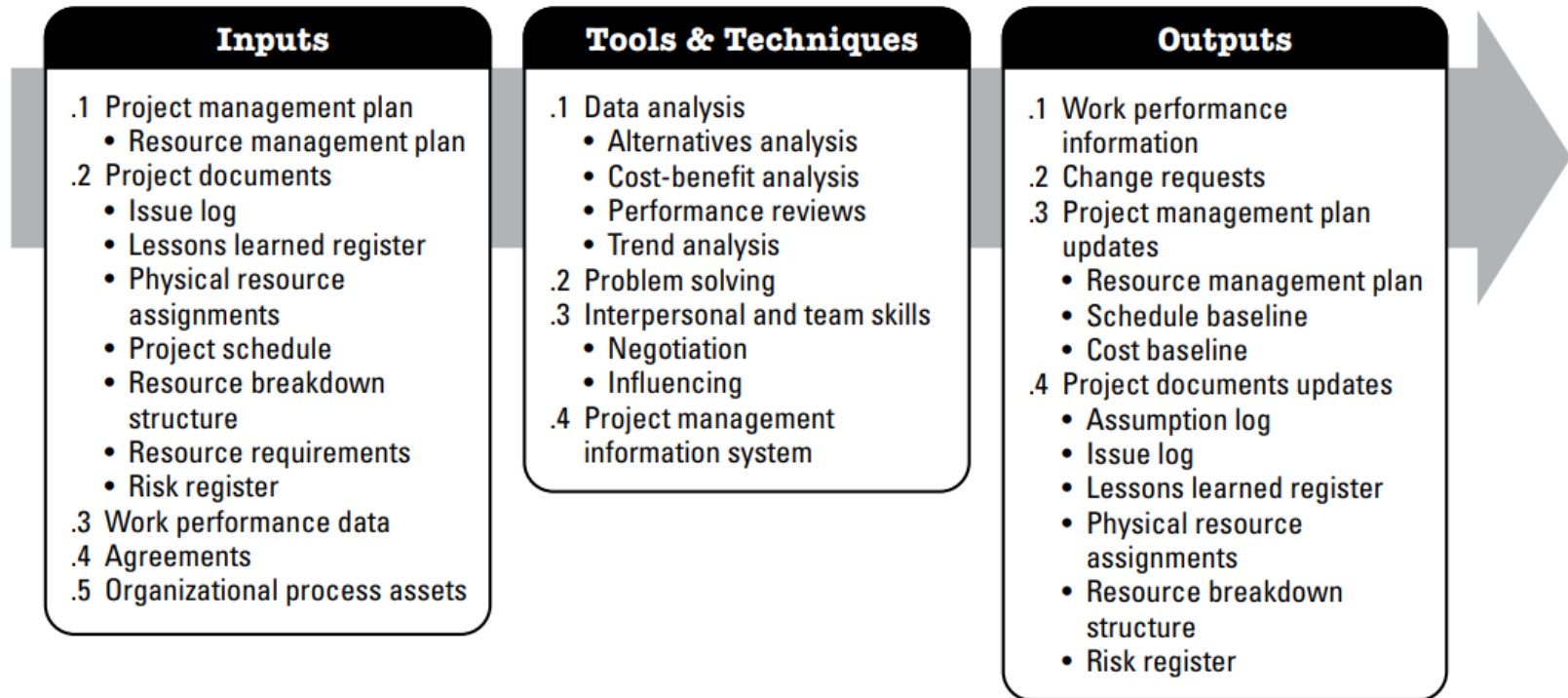
- Ability to be persuasive;
- Clearly articulating points and positions;
- High levels of active and effective listening skills;
- Awareness of, and consideration for, the various perspectives in any situation;
- Gathering relevant information to address issues and reach agreements while maintaining mutual trust

Control Resources

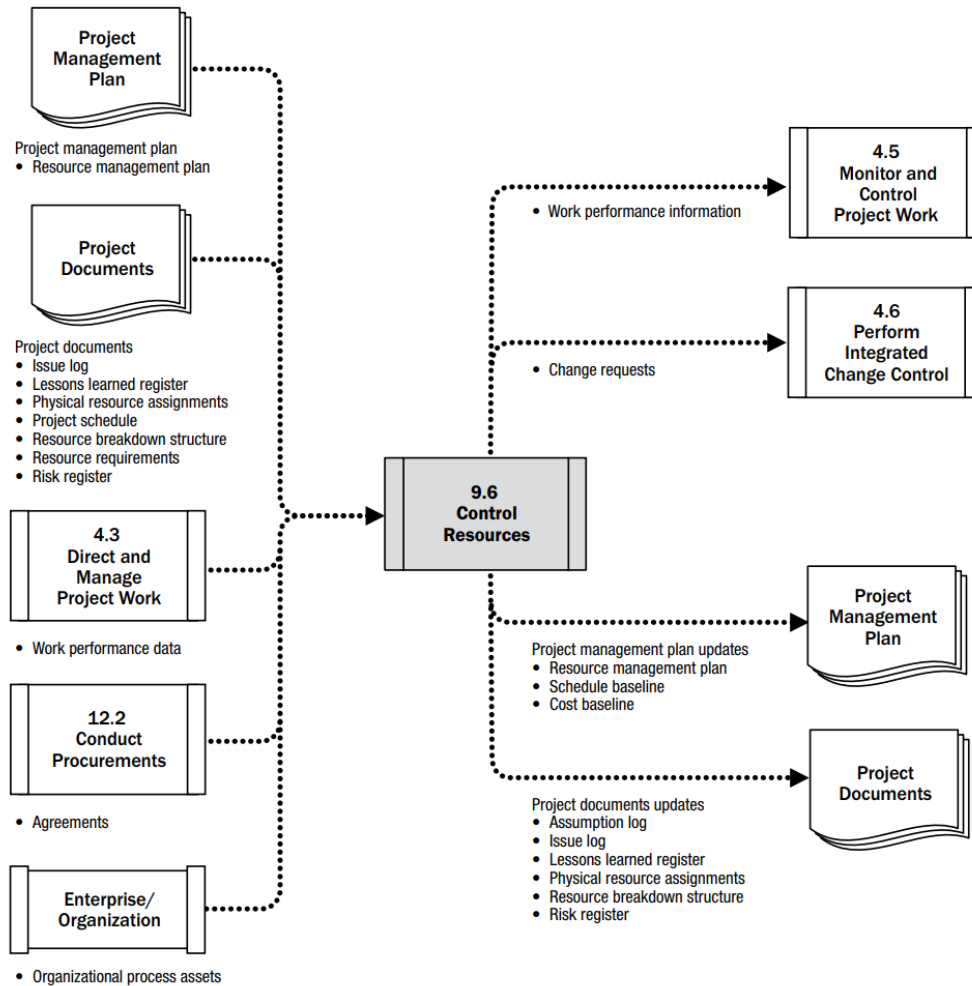
Control Resources

- **What:** process of ensuring that the physical resources assigned and allocated to the project are available as planned, as well as monitoring the planned versus actual utilization of resources and taking corrective action as necessary.
 - The Control Resources process is concerned with physical resources such as equipment, materials, facilities, and infrastructure.
 - Team members are addressed in the Manage Team process
- **Why:** ensuring that the assigned resources are available to the project at the right time and in the right place and are released when no longer needed.
- **When:** performed throughout the project.
- **How:**
 - Monitoring resource expenditures,
 - Identifying and dealing with resource shortage/surplus in a timely manner,
 - Ensuring that resources are used and released according to the plan and project needs,
 - Informing appropriate stakeholders if any issues arise with relevant resources,
 - Influencing the factors that can create resources utilization change
 - Managing the actual changes as they occur

Process



Data Flow



Tools & Techniques: Data Analysis

Data Analysis

Alternatives can be analyzed to select the best resolution for correcting variances in resource utilization.

- Alternatives such as paying additional for overtime or additional team resources can be weighed against a late delivery or phased deliveries.

Cost-benefit analysis.

This analysis helps to determine the best corrective action in terms of cost in case of project deviations.



Tools & Techniques: Data Analysis

Performance reviews

Performance reviews measure, compare, and analyze planned resource utilization to actual resource utilization.



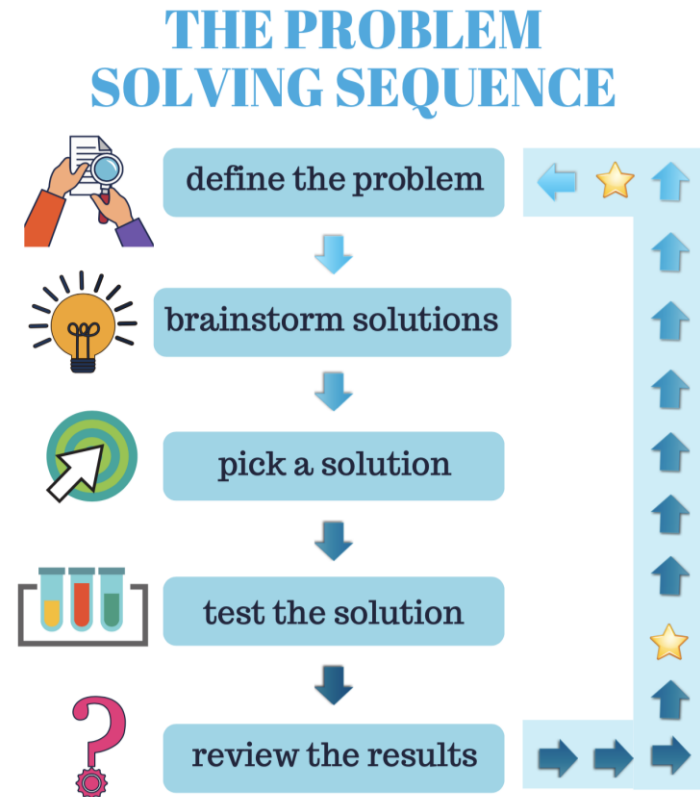
Trend analysis.

Trend analysis examines project performance over time and can be used to determine resources needed at upcoming stages of the project.



Tools & Techniques: Problem solving

- Problem solving may use a set of tools that helps the project manager to solve problems that arise during the control resource process.
 - Inside problem: machines or infrastructure used by another department in the organization and not released in time, materials that have been damaged because of unsuitable storage conditions, etc.
 - Outside problem: major supplier that has gone bankrupt or bad weather that has damaged resources.



Tools & Techniques

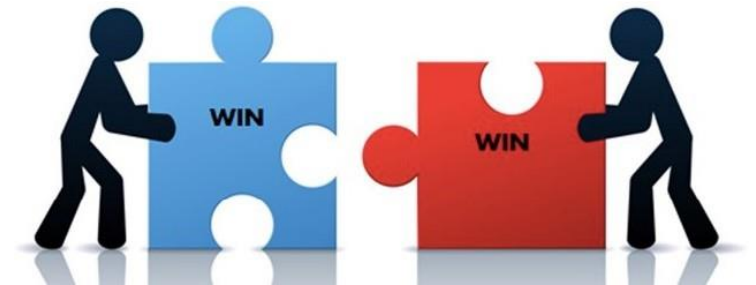
Influencing

Influencing can help the project manager solve problems and obtain the resources needed in a timely manner.



Negotiation

The project manager may need to negotiate for additional physical resources, changes in physical resources, or costs associated with the resources.



Outputs



Project documents updates

Some of the documents that might be affected by the control resources project are: assumption log, issue log, lessons learned register, physical resource assignments, resource breakdown structure, risk register.



Change requests

When you need to make changes to the Resource Management plan, you make them through a change request.



Project Management plan updates

Some of the plans affected by this process are Resource Management plan, schedule baseline, cost baseline..



Work performance information

This is all of the information about how your project is progressing. As you make changes to the resource make-up on your project, those changes are reflected in your work performance information.